

Experiment 1

Question

Perform basic image handling and processing operations on an image. Read an image in Python and convert the image to grayscale using OpenCV.

Aim

To read an image using Python and OpenCV, convert it into a grayscale image, display both the original and grayscale images, and save the grayscale image.

Procedure

1. Import the OpenCV library (cv2).
2. Read the input image using cv2.imread().
3. Check whether the image is loaded successfully.
4. Display the original color image using cv2.imshow().
5. Convert the image into grayscale using cv2.cvtColor().
6. Display the grayscale image.
7. Save the grayscale image using cv2.imwrite().

Code

```
import cv2

# Read image
image = cv2.imread("tree.jpg")

# Check whether image is loaded
if image is None:
    print("Image not found!")
else:
    # Show original image
    cv2.imshow("Original Image", image)

    # Convert to grayscale
    gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

    # Show grayscale image
    cv2.imshow("Grayscale Image", gray)

    # Save grayscale image
    cv2.imwrite("gray_tree.jpg", gray)
```

Input

Input Image: tree.jpg

Output

1. **Original Image** : Displays the original color image of the tree.
2. **Grayscale Image** : Displays the same image converted into grayscale.
3. **Saved File** : gray_tree.jpg is created in the project folder



Result

The image was successfully read using OpenCV, converted into a grayscale image, displayed on the screen, and saved as **gray_tree.jpg**.